GROUP 14

A GENERAL ANOMALY DETECTION FRAMEWORK FOR FLEET-BASED CONDITION MONITORING OF MACHINES

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PROJECT FOCUS:

Early fault detection in machine fleets before failure occurs

KEY INNOVATION:

- Fleet-based approach leveraging similar machines
- Unsupervised anomaly detection without historical labeled data
- Real-time online monitoring with visualization

MEMBERS:

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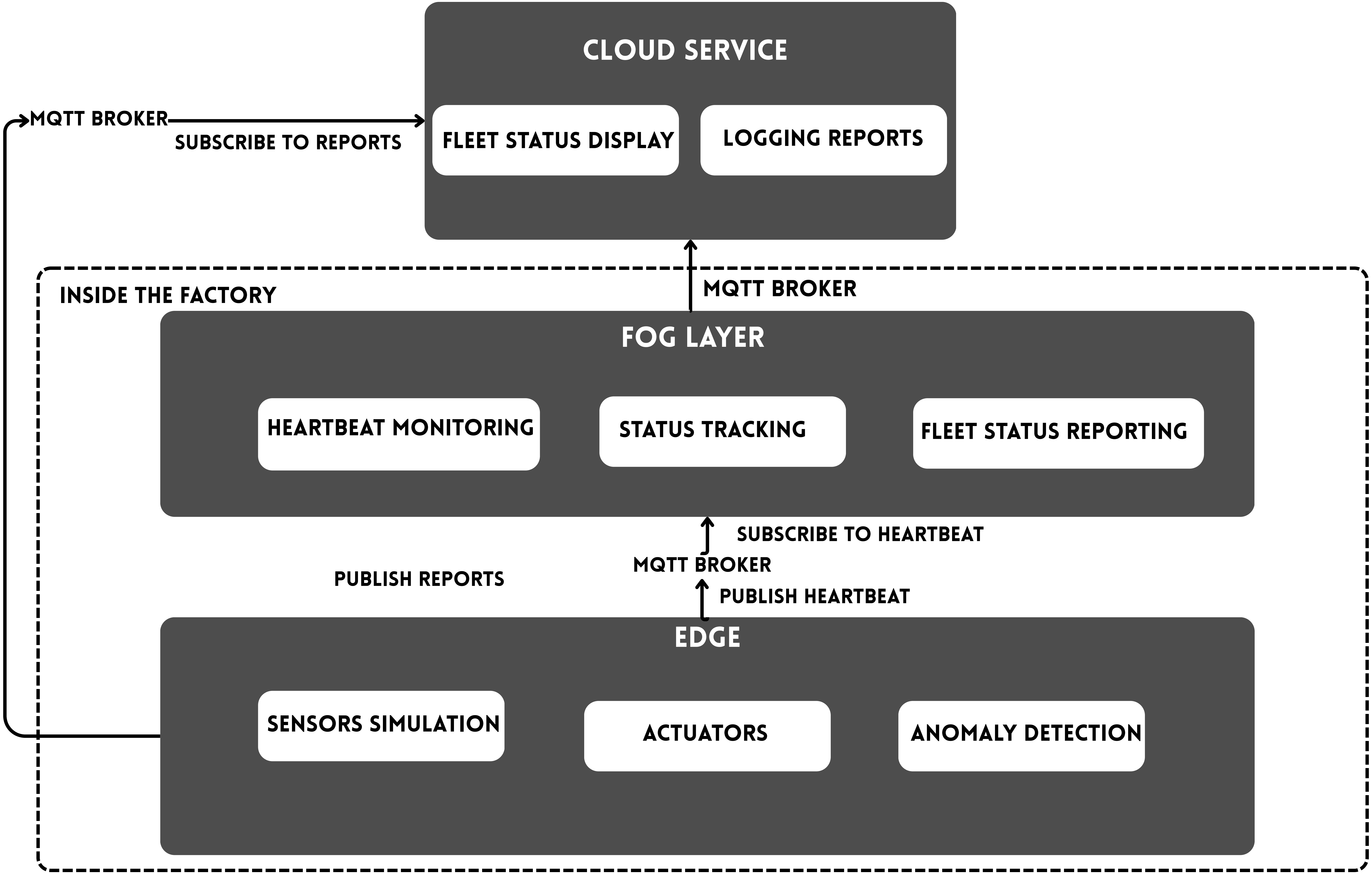
PROBLEM STATEMENT

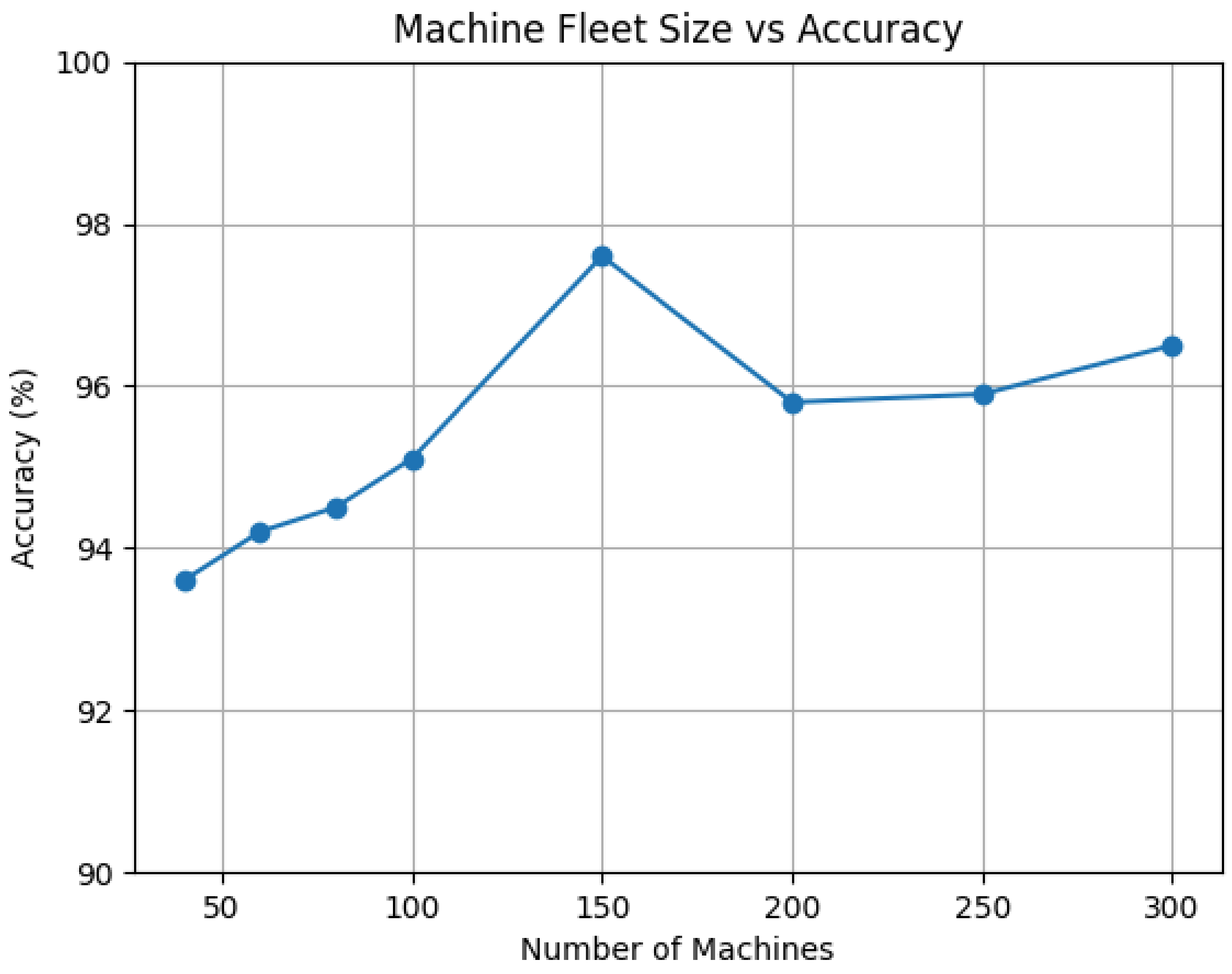
Machine failures decrease up-time and can lead to extra repair costs or even to human casualties. Recent condition monitoring techniques use artificial intelligence in an effort to avoid time-consuming manual analysis and handcrafted feature extraction. Many of these only analyze a single machine and require a large historical data set. In practice, this can be difficult and expensive to collect.

Processing all the data at the cloud will require the data to be transmitted to cloud and then process it, thereby increasing the total latency to identify the fault.

Hence, we designed a edge-fog-cloud architecture that brings the fault-detection to the edge node, and a unsupervised model that removes the need for large historical dataset.


CLOUD SERVICE
FLEET STATUS DISPLAY
LOGGING REPORTS
MQTT BROKER
SUBSCRIBE TO REPORTS
INSIDE THE FACTORY
MQTT BROKER
FOG LAYER
HEARTBEAT MONITORING
STATUS TRACKING
FLEET STATUS REPORTING
SUBSCRIBE TO HEARTBEAT
MQTT BROKER
PUBLISH REPORTS
PUBLISH HEARTBEAT
EDGE
SENSORS SIMULATION
ACTUATORS
ANOMALY DETECTION

ACCURACY

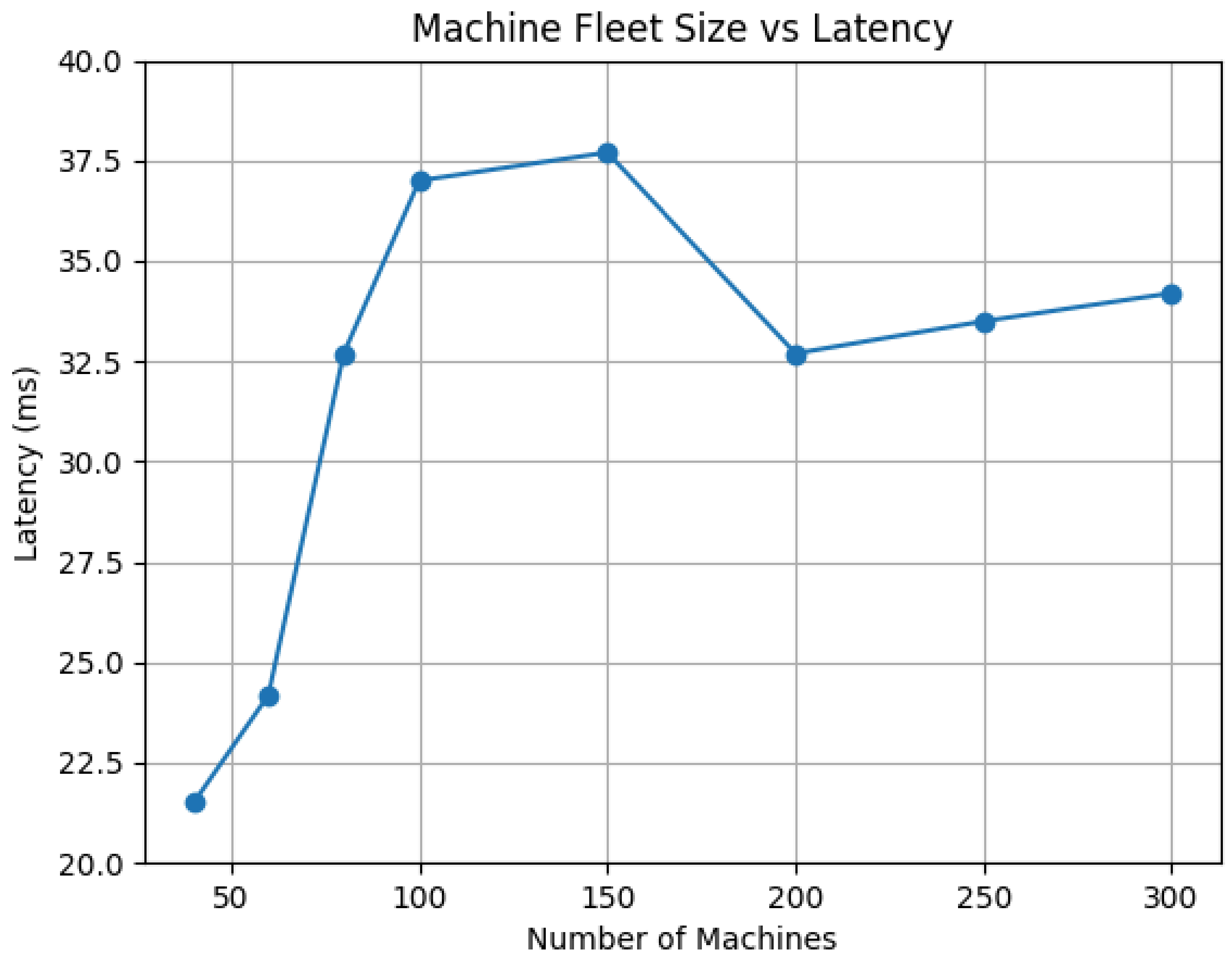

Machine Fleet Size vs Accuracy
Accuracy (%)
100
98
96
94
92
90
50
100
150
200
250
300
Number of Machines

END-END LATENCY

Size of a message for 100 machines = 180KB
Transmition rate of network = 100Mbps

Transmission time required = 14.76ms

(Propagation delay is negligible compared to transmission delay)



PROCESSING POWER

Edge:

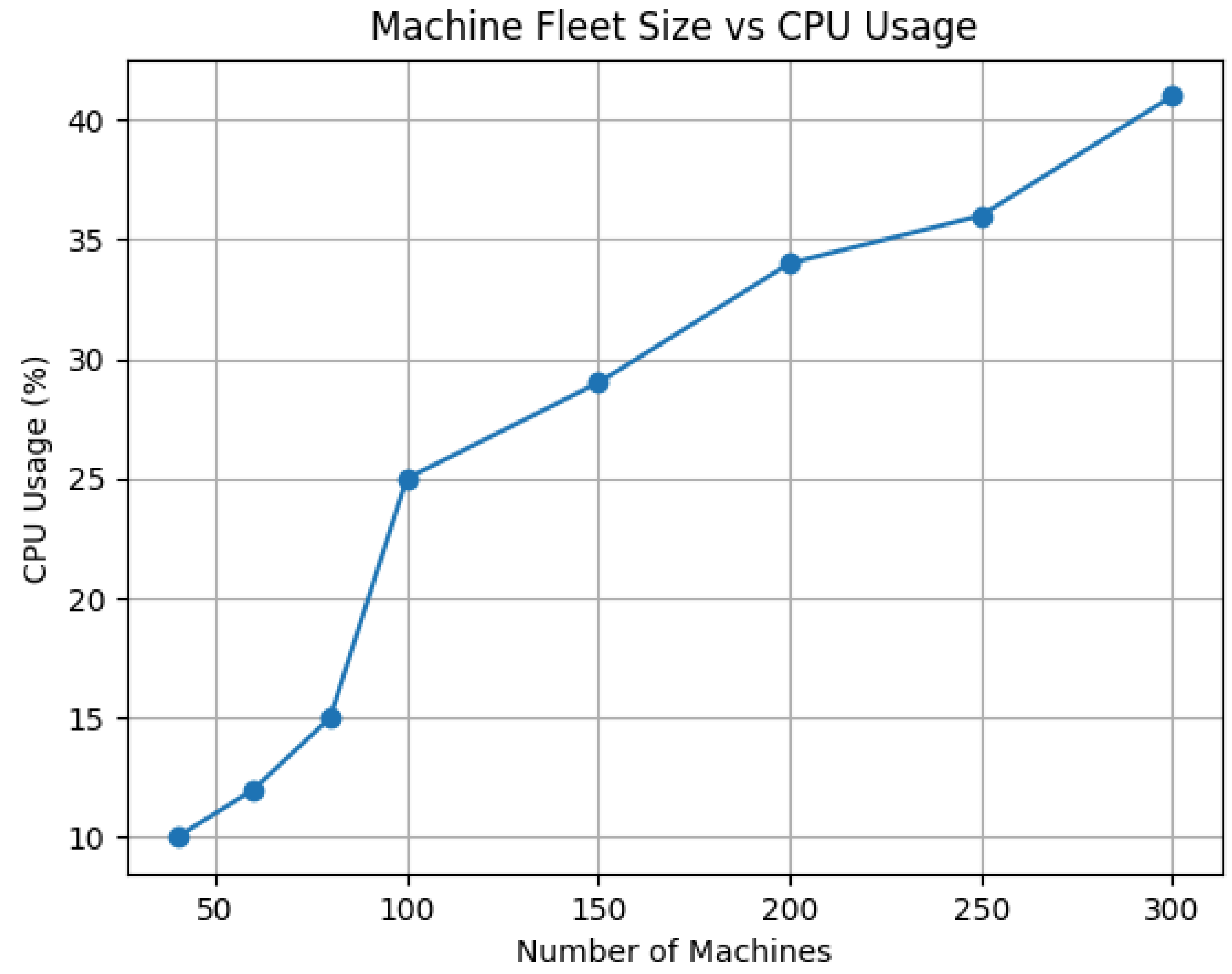
- 60-70MB is consumed by docker.
- 0.5 CPU time of the host machine.

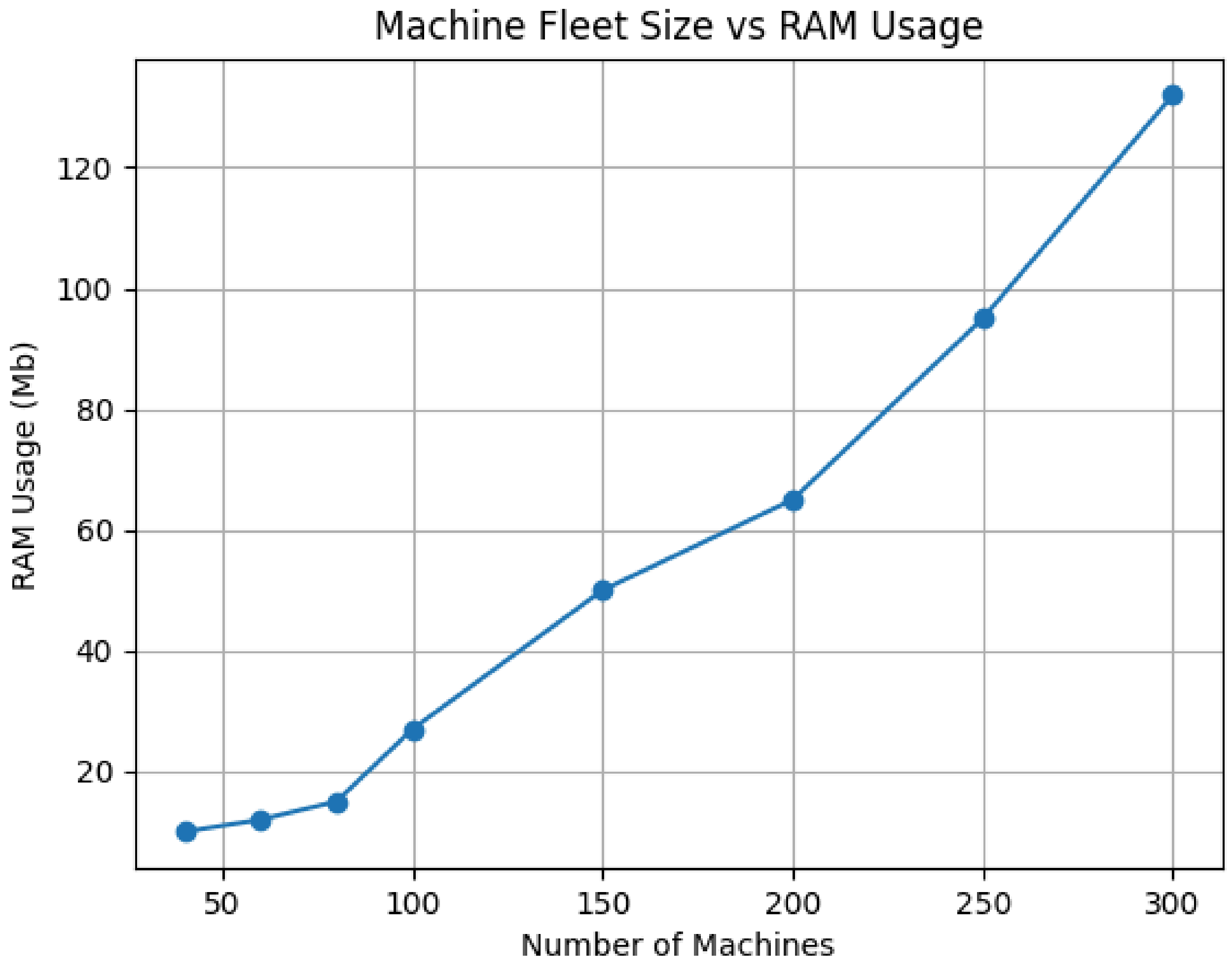
Fog:

- 512MB RAM allocated.
- 1 CPU time of the host machine.

Cloud:

- No limitations on RAM and CPU is imposed.





INDIVIDUAL CONTRIBUTION

Dwarakesh Venkatesh:

- Data generation
- Agglomerative clustering
- Unsupervised Learning

Shyam Sundar Raju:

- Fog layer
- MQTT protocol

Sarigasini:

- Unsupervised Learning
- Edge layer

Pooja Sree:

- Cloud layer
- Heartbeat mechanism